Joint	Fastener	Qty	Grade / finish	Tightening	Notes
Blade root → hub flange	stud M12 × 75 + nut + washer plate	30	10.9, zinc-flake	95 Nm + 90°	Loctite 270 in the insert, 40 Nm
Hub → drum web	M16 × 70 ISO 4017	12	10.9, zinc-flake	250 Nm (μ 0.12)	check after 50 h
Spinner → brackets	M8 × 20 + rubber washer	3	A4-70	12 Nm	
Spindle → bedplate front plate	M20 × 90 ISO 4014	12	10.9	540 Nm	from inside, NP1-100
Stator carrier → spindle	M12 × 45	12	10.9	110 Nm	NP1-200
Brake disc → drum	M12 × 40	12	10.9	110 Nm	NP1-230
Rear cover → drum	M8 × 20	24	A4-80	20 Nm	NS6-610
Caliper → NP1-310	M16 (caliper pattern)	4	10.9	250 Nm	per caliper supplier
NP1-310 → side plate	M16 × 70 + nut	4	10.9 HV	210 Nm	
Lock pin holder → front plate	M12 × 50	2	8.8	80 Nm	
Yaw inner ring → bedplate	M16 × 110 + nut	16	10.9 HV	250 Nm	doc 25
Yaw outer ring → NY-400	M16 × 70	16	10.9	250 Nm	doc 25
Pad unit → bedplate	M16 × 50	8	10.9	250 Nm	NY-410
Gearmotor bracket → side plate	M16 × 60 + nut	8	8.8	195 Nm	NP1-420
Gearmotor → bracket	M10 × 30	8	8.8	49 Nm	
NY-400 → tower top flange	M20 × 90 + nut	16	10.9 HV	450 Nm	through bolts
Cover arms → side plates	M10 × 25	8	A4-70	35 Nm	rubber mounts
Cover lid → tray	M8 × 25 + EPDM	14	A4-70	10 Nm	
Base flange → anchors	M24 nuts (levelling + top + lock)	12	8.8 HDG anchors	450 Nm	after grout cured 7 d
Hinge brackets → anchors	M24 nuts	4	8.8 HDG	350 Nm	
Hand-hole cover	M8 × 20	6	A4-70	12 Nm	EPDM gasket
Cabinet stand → pads	M12 anchors	4	A4-70	40 Nm	

NOTES

1. Torques for lightly oiled zinc-flake threads (μ ≈ 0.12) unless stated; HV sets per EN 14399-4 with the manufacturer's k-class.
2. Mark every torqued bolt with a torque stripe; record values in the installation report. Re-check all structural joints (blade, hub, yaw, tower) after 50 h and then yearly.
3. Never re-use M16 / M20 / M24 HV sets that have been fully tightened; replace studs that show thread damage.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FASTENER AND TORQUE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-900	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
ISO 4014/4017, EN 14399-4 (HV), 10.9 zinc-flake unless stated	–	Claude / not checked